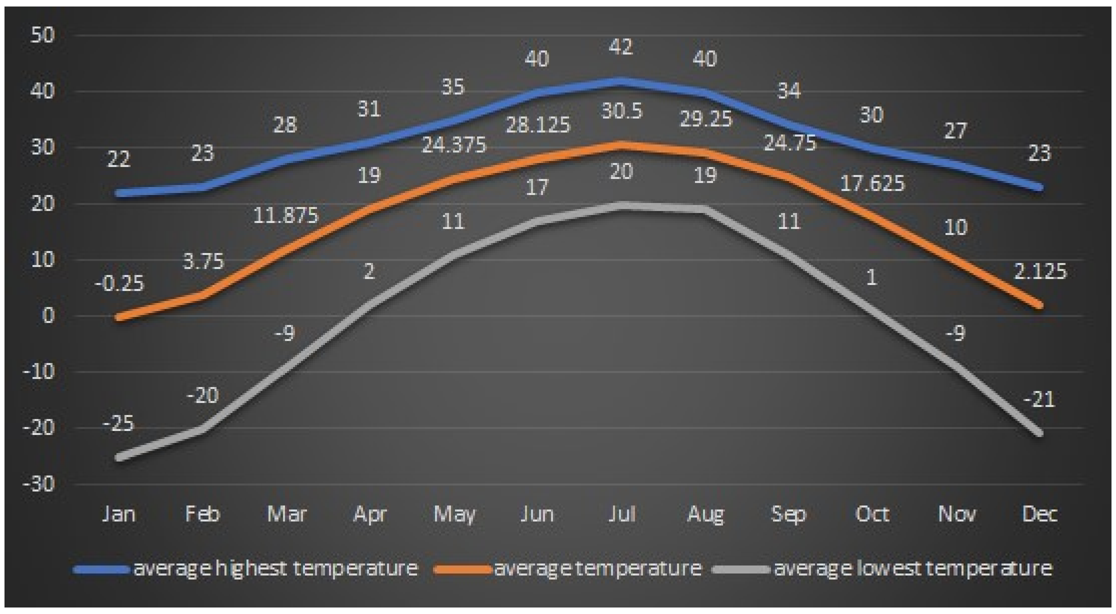
Electrolytic capacitor lifetime estimation

Body temperature-rise method

Average Temperature：



	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Average Highest Temperature	22	23	28	31	35	40	42	40	34	30	27	23
Average Temperature	-0.25	3.75	11.875	19	24.375	28.125	30.5	29.25	24.75	17.625	10	2.125
Average Lowest Temperature	-25	-20	-9	2	11	17	20	19	11	1	-9	-21

Assume：

- The capacitor body temperature gets 85°Cwhen the EST average temperature is 50°C。Temperature-rise is 35°C。
- The capacitor body temperature is changed with EST average temperature. The change rate is the same.

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Average Highest Temperature	22	23	28	31	35	40	42	40	34	30	27	23
core temperature	57	58	63	66	70	75	77	75	69	65	62	58
Average Temperature	-0.25	3.75	11.875	19	24.375	28.125	30.5	29.25	24.75	17.625	10	2.125
core temperature	34.75	38.75	46.875	54	59.375	63.125	65.5	64.25	59.75	52.625	45	37.125
Average Lowest Temperature	-25	-20	-9	2	11	17	20	19	11	1	-9	-21
core temperature	10	15	26	37	46	52	55	54	46	36	26	14

SPEC LIST FOR CD29H SERIES

Part Number	Description	WV (V)	CAP (uF)	tg δ (%)	LC (mA/5min)	Rated ripple (Arms/120Hz, 105℃)	D×L
ECS2WQH391MLA250050	CD29H-450V-390uF-25*50	450	390	12	1.5	1.67	25x50

Ripple current caculation method

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Average Highest Temperature	22	23	28	31	35	40	42	40	34	30	27	23
Actual operation temperature	52	53	58	61	65	70	72	70	64	60	57	53
Average Temperature	-0.25	3.75	11.875	19	24.375	28.125	30.5	29.25	24.75	17.625	10	2.125
Actual operation temperature	29.75	33.75	41.875	49	54.375	58.125	60.5	59.25	54.75	47.625	40	32.125
Average Lowest Temperature	-25	-20	-9	2	11	17	20	19	11	1	-9	-21
Actual operation temperature	5	10	21	32	41	47	50	49	41	31	21	9

Assume：

- The actual operation temperature gets 50°C when the EST temperature gets 80°C（body temperature85°C－△T=5°C）
- The actual operation temperature is changed with EST temperature. The change rate is the same.

Upper category temperature: $T_0 := 105\ °C$

Ambient temperature in the application:
(Calculate at maximum ambient temperature) $T_A := [65\ °C\ 65\ °C\ 65\ °C\ 65\ °C\ 65\ °C\ 65\ °C\ 65\ °C\ 65\ °$

Lifetime at rated ripple and at rated temperature $L_0 := 5000 \cdot hr$

Ripple current in the application: $I_A := 4.2 \cdot A$

Normal ripple current at upper category temperature: $I_R := 2.85 \cdot A$

Core temperature increase of the elcap:
（typ.3,5-5K at T0=105℃&3,5-10K at T0=85℃） $\Delta T := 5 \cdot K$

Empirical safety factor, defined as:

$K_a := 2$

$$K_b := \begin{cases} 2 & \text{if } I_A > I_R \\ 4 & \text{if } I_A \le I_R \end{cases}$$

$$K_i := \begin{cases} K_b & \text{if } T_0 = 105\ °C \\ K_a & \text{if } T_0 = 85\ °C \end{cases}$$

Rated votage: $U_R := 450 \cdot V$

Actual operating votage: $U_A := 400 \cdot V$

Exponent defined as:
(Operating votages below half rated votage are considered impractical and are not covered by the model.JiangHai has intentionally chosen an exponent of 2.5 thus offering a moderate estimation) $n := 2.5$

Structure of life time model:

$$L_x = L_0 \cdot K_T \cdot K_R \cdot K_V$$

Where:

Lx Estimated Lifetime

L0 Lifetime at rated ripple and at rated temperature

KT Temperature factor(ambient temperature)

KR Ripple current factor(self-heating)

KV Operating votage

Temperature factor KT :

The lifetime of elcaps follows the industry wide well established "10-Kelvin-rule" from Arrhenius:a drop of the temperature by 10 K doubles the lifetime[1,3,4,6,9].The formula for KT reads:

$$K_T := 2^{\left(\frac{T_0 - T_A}{10 \cdot K}\right)^{T}}$$

Ripple current KR :

Jianghai estimates the impact of the applied ripple current on the self-heating and in ture on elcap lifetime by following formula:

$$K_R := \boxed{K_i}^{\left(1 - \left(\frac{I_A}{I_R}\right)^2\right) \cdot \frac{\Delta T}{10 \cdot K}}$$

Voltage factor KV

Operating votages below half rated votage are considered impractical and are not covered by the model.JiangHai has intentionally chosen an exponent of 2.5 thus offering a moderate estimation.

$$K_V := \begin{cases} \left(\frac{U_A}{U_R}\right)^{-n} & \text{if } 0.6 \le \left(\frac{U_A}{U_R}\right) \le 1 \\ 3.59 & \text{if } 0 \le \left(\frac{U_A}{U_R}\right) \le 0.6 \end{cases}$$

The average time coefficient of one year is as follows:

$$K_{Tavg} := \boxed{mean\left(\boxed{K_T}\right)} = ?$$

The capacitance life is estimated as follows:

$$L_X := L_0 \cdot \boxed{K_{Tavg}} \cdot K_R \cdot K_V = ?\ hr$$